

TERESA PANURACH

U.S. Naval Research Laboratory, 4555 Overlook Ave SW, Washington, DC 20375

RESEARCH INTERESTS

surveys, globular clusters, low-mass X-ray binaries, radio astronomy, high energy phenomena

APPOINTMENTS

U.S. Naval Research Laboratory , Washington D.C. National Research Council Postdoctoral Associate	2026 - Present
Norfolk State University (NSU) , Norfolk, VA Executive Director of Education, CREST	2025 - 2026
Norfolk State University , Norfolk, VA Executive Director, NoVEL Consortium	2023 - 2025

EDUCATION

Michigan State University (MSU) , East Lansing, MI Ph.D in Astrophysics and Astronomy	2023
Michigan State University , East Lansing, MI M.S in Astrophysics and Astronomy	2021
The City University of New York (CUNY) at Hunter College , New York, NY B.A in Physics	2018

AWARDS, GRANTS, AND FELLOWSHIPS

HONES Michigan Space Grant Consortium: Discovering Astronomy with LSST: Resources to Promote Research, PIs: E. Cackett, K. Dage & T. Panurach , \$5000	2024
Alliances with Under-Resourced Institutions LSST Discovery Alliance Inclusive Collaboration, PIs: K. Dage & T. Panurach , \$30,000	2023
Packard Foundation: Increasing Diversity in STEM, Co-PI: T. Panurach , \$85,3000	2023
#iteachmsu Educator Award	2021
NSF Graduate Research Fellowship, \$138,000	2020 - 2023
AstroCom NYC Fellowship, Full Tuition Coverage and Summer Research Stipend	2016 - 2018
American Astronomical Society FAMOUS Travel Grant (229th Meeting), \$500	2018
AstroCom NYC Senior Scholarship, \$700	2017

ACCEPTED TELESCOPE PROPOSALS

<i>Radio Monitoring of Her X-1 During its Superorbital Period</i> 12 hours, PI: T. Panurach	VLA 24B-265, VLA 26A-145
<i>A Radio Survey Ultra-Compact X-ray Binaries</i> 90 hours, PIs: K. Dage & T. Panurach	ATCA CS024
<i>Radio Observations of Neutron Star Ultra-Luminous X-ray Sources</i> 12 hours, Co-PI: T. Panurach	ATCA C3547
<i>Tracking the Enigmatic Globular Cluster Ultracompact X-ray Binary X1850-087</i> 3 hours, Joint with Swift (6ks), PI: T. Panurach	VLA 22A-111

TEACHING

Education Coordinator

PHY 241/APS 110: Physics Seminar (12 students, NSU) Fall 2025, Spring 2026

Education Coordinator

PHY 399: Advanced Laboratory (1 student, NSU) Fall 2025

Lead Instructor

Introduction to Python Coding (7 students, NSU) Spring 2025

Teaching Assistant

AST 208: Planets and Telescopes (28 students, MSU) Spring 2020

Teaching Assistant

ISP 205L: Visions of the Universe (6 sections, 600 students, MSU) Fall 2018, Spring 2019, Fall 2019

STUDENT RESEARCH SUPERVISION

Research

Payton Brown (NSU, AIP TEAM-UP Scholar) 2024 - Present

Samuel Uwechue (NSU) 2025

Kavon Ragland (Virginia State U.) 2024 - 2025

Ethan Agudzi-Addo (NSU) 2024 - 2025

Jennifer Rodriguez (MSU, Charles Drew Scholar–grad school at Ohio State U.) 2019 - 2020

Jerry Ortiz (CUNY at College of Staten Island, AstroCom NYC Scholar) 2019

Career

Dion Jordan (NSU, NoVEL SURE Scholar) 2025

Brooke Richardson (NSU, NoVEL SURE Scholar) 2025

Lawrence Tyler (Virginia State U., NoVEL SURE Scholar) 2025

Tyon Phillips (Elizabeth City State U., NoVEL SURE Scholar) 2025

Zakyan Dunn (Elizabeth City State U., NoVEL SURE Scholar) 2025

Jared Mims (Elizabeth City State U., NoVEL SURE Scholar) 2025

Shakiya Yindui (NSU, NoVEL SURE Scholar) 2024

De'ja Bailey (NSU, NoVEL SURE Scholar) 2024 - 2025

Alice Cuffee (NSU, NoVEL SURE Scholar) 2024 - 2025

Jaycie Jenkins (NSU, NoVEL SURE Scholar) 2024 - 2025

Kavon Ragland (Virginia State U., NoVEL SURE Scholar) 2024 - 2025

Chantel Johnson (NSU, NoVEL SURE Scholar) 2023 - 2025

Jordan Simmons (NSU, NoVEL SURE Scholar) 2023 - 2025

Shamaar Howard (NSU, NoVEL SURE Scholar) 2023 - 2025

Curtis Jordan (NSU, NoVEL SURE Scholar) 2023 - 2025

Asia Jones (NSU, NoVEL SURE Scholar) 2023 - 2025

Elias Anwar (NSU, NoVEL Fellow) 2023 - 2025

Ashleigh Wilson (NSU, NoVEL Fellow) 2023 - 2025

Jaylin Coleman (NSU, NoVEL SURE Scholar) 2023 - 2024

Anthony Martin (Virginia State U., NoVEL SURE Scholar) 2023 - 2024

Dominic Ross (Virginia State U., NoVEL SURE Scholar) 2023 - 2024

Miguel Montalvo (UC Berkley, NAC Fellow– grad school at Princeton U.) 2022 - 2023

Daniel Gallego (CUNY City Tech, NAC Fellow) 2022 - 2023

Miranda Pikus (MSU, Charles Drew Scholar–grad school at Purdue U.) 2022 - 2023

Payton Walton (MSU, Charles Drew Scholar) 2022 - 2023

Atticus Chong (MSU, Charles Drew Scholar) 2021 - 2023

Ryan Copeland (MSU, Charles Drew Scholar) 2021 - 2023

Lauren McDermott (MSU, Charles Drew Scholar–grad school at Washington State U.) 2021 - 2022

Elias Taira (MSU, Charles Drew Scholar) 2021 - 2023

Anita Agasaveeran (MSU, Charles Drew Scholar) 2021 - 2023

Ashley Stone (MSU–grad school at West Virginia U.)	2021 - 2023
Wasundara Athukoralalage (MSU, SURF Fellow at Caltech–grad school at Harvard U.)	2020 - 2023

CONFERENCE AND SEMINAR TALKS

Hampton University Physics Colloquium (Norfolk, VA)	Oct 2025
Old Dominion University Physics Colloquium (Norfolk, VA)	Sept 2025
Norfolk State University Physics Seminar (Norfolk, VA)	March 2023
241st American Astronomical Society Dissertation Talk (Seattle, WA)	Jan 2023
National Society of Black Physicists Contributed Talk (Charlottesville, VA)	Nov 2022
Liverpool John Moores University Journal Club Speaker (Virtual)	Sept 2022
MSU Astronomy Seminar (Virtual)	April 2022
Northwestern University CIERA Science Happy Hour (Virtual)	April 2022
American Museum of Natural History Astronomy Colloquium (Virtual)	Jan 2022
Compact Objects 9 (Virtual)	May 2021
Vanderbilt University Astronomy Seminar (Virtual)	Nov 2020
Chandra Frontiers in Time-Domain Science Lightning Talk (Virtual)	Oct 2020
MSU Physics Graduate Organization Seminar (Virtual)	Sept 2020
AAS: High Energy Astronomy Division Frontier Seminar (Virtual)	Sept 2020
REU Symposium (New York, NY)	Aug 2017
CUNY at Hunter College Undergraduate Research Conference (New York, NY)	April 2017
Society for Physics and Astronomy Research Conference (New York, NY)	Oct 2016
REU Symposium (New York, NY)	Aug 2016

OUTREACH AND PUBLIC TALKS

STARS Science Career Panel (Norfolk, VA)	July 2023
Intrepid Sea, Air and Space Museum’s GOALS for Girls Presenter (Virtual)	July 2021
MSU Twitter Manager (@MSU_astro)	2021 - 2023
One Million Women in STEM Featured Scientist	Jan 2021
Intrepid Sea, Air and Space Museum’s GOALS for Girls Presenter (Virtual)	July 2020
MSU’s Science Festival Presenter (East Lansing, MI)	April 2019
Astronomy on Tap: Lansing Presenter (Lansing, MI)	Sept 2018
Astronomy on Tap: New York City Presenter (Brooklyn, NY)	June 2018

CONFERENCE POSTERS

235th American Astronomical Society Meeting (Honolulu, HI)	Jan 2020
231st American Astronomical Society Meeting (Washington D.C)	Jan 2018
Columbia University Astrofest (New York, NY)	Sept 2017
229th American Astronomical Society Meeting (Grapevine, TX)	Jan 2017
Princeton University Conference for Undergraduate Women in Physics (CUWiP)	Jan 2017

ACADEMIC SERVICE

AIP Society of Physics Students: NSU Chapter Faculty Advisor	2025 - Present
Time/Funding Allocation Committees: NASA ADAP (Panel Chair), NASA GO Reviewer	2023 - Present
Publication Referee: MNRAS, ApJ	2023 - Present
MSU Astronomy Group Seminar Committee	2022 - 2023
MSU Astronomy Group Website Committee	2021 - 2023
P-A Research Experiences with Drew Scholars Conference Organizer	2021 - 2022
Physics and Astronomy Department Colloquium Host	2020 - 2021
Co-creator and organizer of Astro Coffee Journal Club	2020 - 2021

X-ray Binary Journal Club Host	2020 - 2021
Graduate Student Search Committee Representative for Astronomy Faculty Position	2020
P-A Research Experiences with Drew Scholars Junior Director	2020 - 2023
National Radio Consortium Chapter Leader	2019 - 2023
Graduate Student Recruitment Organizer	2019
CUWiP Local Organizing Committee	2018 - 2019

PROFESSIONAL DEVELOPMENT WORKSHOPS

Mentoring Relationships and Owning Your Career (Virtual, Henry Ford College)	June 2024
Stellar Mentorship Program Mentor (East Lansing, MI)	2020 - 2023
Maximizing Your Mentoring Relationships Workshop (East Lansing, MI)	June 2020
P-A Research Experiences with Drew Scholars Junior Director (East Lansing, MI)	2020 - 2023
National Radio Consortium Chapter Leader (East Lansing, MI)	2019 - 2023
Maximizing Your Mentoring Relationships Workshop (Charlottesville, VA)	June 2019
Maximizing Your Mentoring Relationships Workshop (New York, NY)	June 2019
Maximizing Your Mentoring Relationships Workshop (East Lansing, MI)	June 2019
Applying to REU and Internships Workshop (MSU CUWiP)	Jan 2019
Maximizing Your Mentoring Relationships Workshop (New York, NY)	June 2018
Maximizing Your Mentoring Relationships Workshop (New York, NY)	June 2017

SOFTWARE KNOWLEDGE AND SKILLS

Technologies: Python (pandas, matplotlib, numpy, scipy, scikit-learn, astropy), Jupyter-Notebook/Lab, Anaconda, UNIX/Linux, Git, statistics, statistical modeling, regression analysis
Major Astronomical Packages: AstroPy, CASA, IRAF
Languages: English (native), Thai (fluent)

PUBLICATION LIST

Refereed Publications:

16. *Radio Monitoring of Hercules X-1*

T. Panurach et al. in prep.

15. *Classifying Compact Radio Emission in Nearby Galaxies: a 10GHz Study of Active Galactic Nuclei, Supernovae, Anomalous Microwave Emission and Star Forming Regions*

K.C. Dage et al. (including **T. Panurach**), 2025, Accepted to the Astrophysical Journal

14. *Observational Astrophysics Research through Cross Institutional Research Partnerships*

K.C. Dage et al. (including **T. Panurach**), 2025, Accepted to The Physics Teacher

13. *Radio Continuum Studies of Ultra-Compact and Short Orbital Period X-Ray Binaries*

K.C. Dage, **T. Panurach** et al., 2025, Astrophysical Journal, 988, 131

12. *PSR J19471120: A New Huntsman Millisecond Pulsar Binary*

J. Strader et al., (including **T. Panurach**), 2025, Astrophysical Journal, 980, 124

11. *Detecting the Black Hole Candidate Population in M51s Young Massive Star Clusters: Constraints on Accreting Intermediate Mass Black Holes*

K.C. Dage et al., (including **T. Panurach**), 2024, Astrophysical Journal, 979, 82

10. *Do Neutron Star Ultra-Luminous X-Ray Sources Masquerade as Intermediate Mass Black Holes in Radio and X-Ray?*

T. Panurach et al., 2024, Astrophysical Journal, 977, 2

9. *A Survey for Radio Emission from White Dwarfs in the VLA Sky Survey*
I. Pelisoli et al., (including **T. Panurach**), 2024, Monthly Notices of the Royal Astronomical Society, 531, 1805P
8. *Revisiting the Classics: On the Evolutionary Origin of the Fe II and He/N Spectral Classes of Novae*
E. Aydi et al., (including **T. Panurach**), 2024, Monthly Notices of the Royal Astronomical Society, 527, 9303A
7. *Catching a Nova X-ray/UV Flash in the Visible? Early Spectroscopy of the Extremely Slow Nova Velorum 2022 (Gaia22alz)*
E. Aydi et al., (including **T. Panurach**), 2023, Monthly Notices of the Royal Astronomical Society, 524, 1946A
6. *Tracking the Enigmatic Globular Cluster Ultracompact X-ray Binary X1850–087: Extreme Radio Variability in the Hard State*
T. Panurach et al., 2023, Astrophysical Journal, 946, 88P
5. *Multi-wavelength Observations of a New Redback Milisecond Pulsar Candidate 4FGL J1910.75320*
K. Au et al., (including **T. Panurach**), 2022, Astrophysical Journal, 943, 103A
4. *The MAVERIC Survey: Variable Jet–Accretion Coupling in Luminous Accreting Neutron Stars in Galactic Globular Clusters*
T. Panurach et al., 2021, Astrophysical Journal, 923, 88
3. *GS2000+25: The Least Luminous Black Hole X-ray Binary*
J. Rodriguez et al., (including **T. Panurach**), 2020, Astrophysical Journal, 889, 58R
2. *Constraints On Blue Straggler Formation Mechanisms in Galactic Globular Clusters from Proper Motion Velocity Distributions*
N. W. C. Leigh, **T. Panurach**, et al., 2019, Monthly Notices of the Royal Astronomical Society, 482, 231
1. *When Do Clusters Become Multiple Star Systems?: II. Towards a Half-Life Formalism with Four Bodies*
T. Ibragimov et al., (including **T. Panurach**), 2018, Monthly Notices of the Royal Astronomical Society, 477, 4213

Other Publications:

8. [NoVEL Consortium Official Website](#) June 2023
T. Panurach. NSU College of Science, Engineering, and Technology: Programs.
7. [The Astronomy Group Welcomes Two Graduate Students](#) Dec 2022
T. Panurach. MSU Astronomy Group Newsletter.
6. [Unforecasted Turbulence: Teresa Panurach](#) May 2022
T. Panurach. MSU College of Natural Science: NatSci Profiles (Reposted from Astrobit.es).
5. [Celebrating this Year's Milestones](#) May 2022
T. Panurach. MSU Astronomy Group Newsletter.
4. [Unforecasted Turbulence: Teresa Panurach](#) May 2022

T. Panurach. Astrobites.

3. *The Astronomy Group Welcomes Three Graduate Students* Jan 2022

T. Panurach. MSU Astronomy Group Newsletter.

2. *Undergraduate Jessie Miller Wins Thomas Osgood Outstanding Senior Award* June 2021

T. Panurach. MSU Astronomy Group Newsletter.

1. *Four Astronomy Undergraduates Receive Lawrence W. Hantel Endowed Fellowship* March 2021

T. Panurach. MSU Astronomy Group Newsletter.